Eclipse Install for Windows with MinGW

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Purpose

Provide guidance for installing the MinGW C++ compiler and Eclipse on Windows. Please note that these instructions will age out of date quickly. I recommend that you always install the latest version of any package listed in these instructions.

Java

The Eclipse environment is built using Java. In order for Eclipse to run, you must have Java properly installed.

Java comes in two main forms: the java runtime package and the java development kit package

Java comes in both 32 and 64 bit versions.

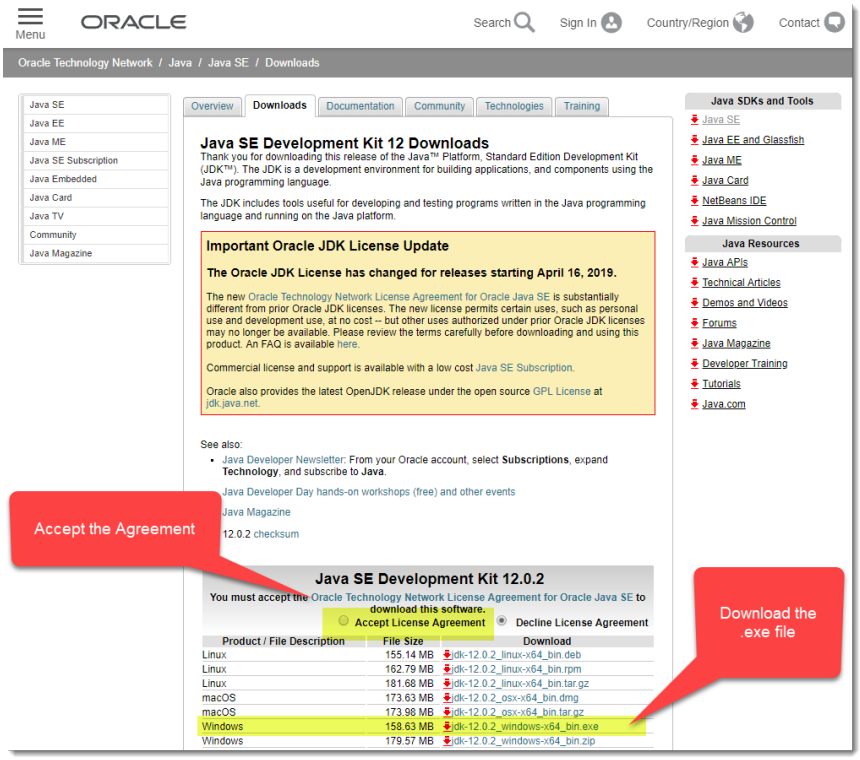
I recommend that you install the 64-bit java development kit.

Download Java

Download java from Oracle here:

https://www.oracle.com/technetwork/java/javase/downloads/index.html

Select the most recent version of the "Java Platform, Standard Edition"



Install Java

Run the installer that you downloaded. Do not make any changes to the locations and select the next button until installation is complete. You will see a screen like this:

Close the installer

Add the Java JDK to the "Path" Environment Variable

This is where many people fail to get things working properly. I found a Youtube video that gives the steps. Please pay attention to how you do this because you will need to repeat steps similar to this later when you install MinGW.

https://www.youtube.com/watch?v=uadGsNA6h5Q

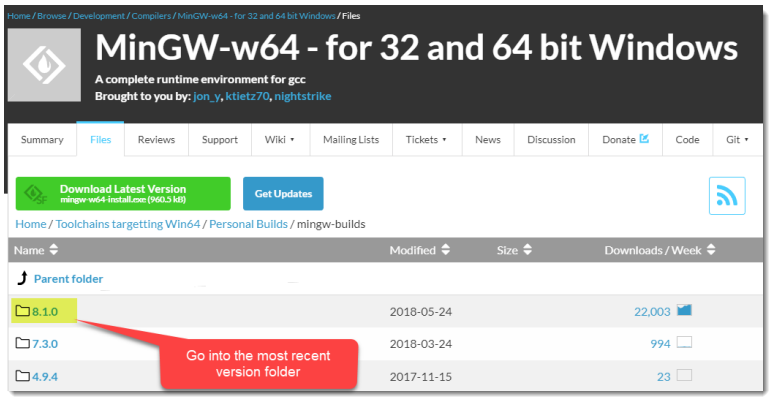
MinGW

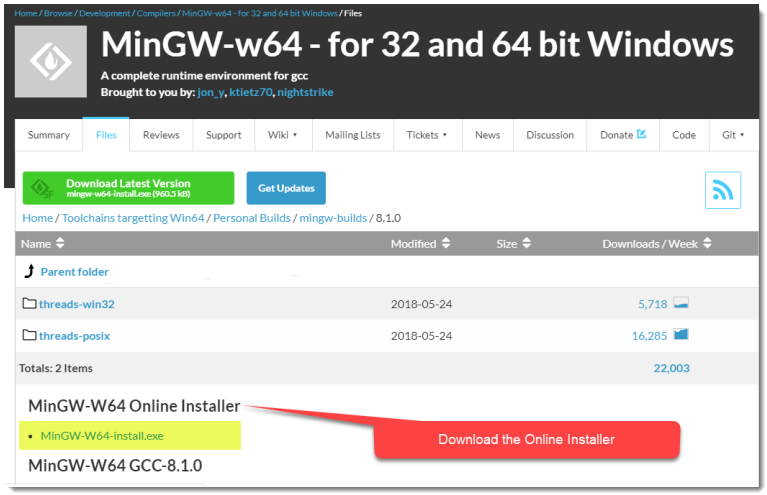
MinGW is a Microsoft Windows port of the GNU C++ compiler. Since Eclipse is a development environment but it does not include a compiler, we must install the compiler prior to installing Eclipse.

Download MinGW

Please download the most recent version of the MinGW installer from here:

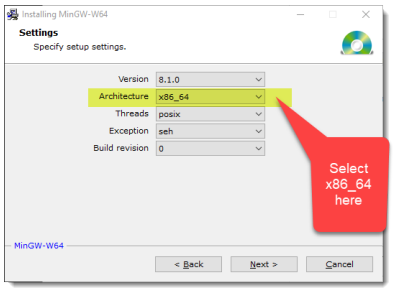
https://sourceforge.net/projects/mingw-w64/files/Toolchains%20targetting%20Win64/Personal%20Builds/mingw-builds/





Install MinGW

Run the file you just downloaded. On the second installer screen, make sure to select the X86_64 architecture. Don't make any other changes with the installer.



Add the MinGW compiler the "Path" Environment Variable

This is where many people fail to get things working properly. Refer back to the Java install where you added to the "Path" environment variable. Find the path for MinGW and copy it to you copy buffer the same way you did for Java. The path will look similar to this:

C:\Program Files\mingw-w64\x86_64-8.1.0-posix-seh-rt_v6-rev0\mingw64\bin

As before, open the Environment Variable area and add this to the "Path" environment variable.

Testing the MinGW compiler

To test that your install was successful, please close any Windows Command Prompt windows that you have open. Then, open a brand new Command Prompt. Enter this command:

```
g++ --version
```

You should see output similar to the following:

```
g++ (x86_64-posix-seh-rev0, Built by MinGW-W64 project) 8.1.0
Copyright (C) 2018 Free Software Foundation, Inc.
This is free software; see the source for copying conditions.  There is NO
warranty; not even for MERCHANTABILITY or FITNESS FOR A PARTICULAR PURPOSE.
```

If you get an error message telling you that g++ is not recognized as an internal or external command, you need to make sure you have added to your "Path" environment variable.

You can close the command prompt by typing in the following command:

```
exit
```

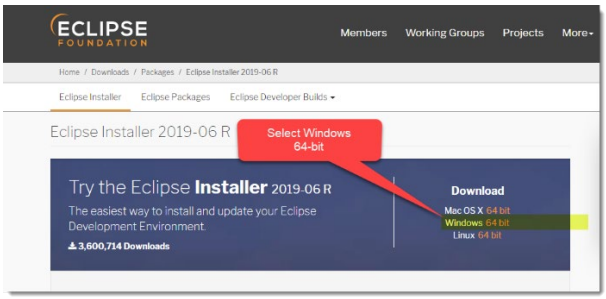
Eclipse

Once you have successfully installed the Java SE SDK and the MinGW compiler, it is time to download and install Eclipse. Eclipse is distributed in many different variations. We want to download the Eclipse that is pre-configured to support C/C++ developers. This variation is called "Eclipse CDT (the C/C++ support for Eclipse)." You will want to download the latest version on the website.

Download Eclipse

Eclipse now comes with a proper installer. Download the latest installer from here:

https://www.eclipse.org/downloads/packages/installer



Install Eclipse

Run the installer. Make sure to select the C/C++ version:

Accept any/all licenses and do not make any other changes. When complete, you should be able to launch Eclipse.